TEXT EXTRACTION FROM AN IMAGE

Advanced Computer Vision and Image Processing

Submitted by

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CONTENTS

List of Acronyms **vii**

Abstract **1**

1 Introduction **3**

1.1 Introduction 3

1.2 Background 3

1.3 Problem Statement 3

1.4 Aim and Objectives 4

1.5 Research Questions 4

1.6 Significance of the Study 4

Summary of the Chapter 4

2 Literature Review **6**

2.1 Introduction 6

2.2 Traditional OCR Systems 6

2.3 Scene Text Detection Approaches 7

2.4 Scene Text Recognition Methods 8

2.5 End-to-End Text Extraction Pipelines 8

Summary of the Chapter 9

3 Research Methodology **10**

3.1 Introduction 10

3.2 Review of Related Research Work 10

3.3 Study of Current Techniques 10

3.4 Study and Examination of Different Approaches 10

3.5 Development of New Algorithm 11

3.6 Description of Proposed System Design 11

3.6.1 System Overview 11

3.6.2 Preprocessing Stage 14

3.6.3 Text Detection Stage 14

3.6.4 Text Recognition Stage 14

3.6.5 Output and Storage 14

3.6.6 System Advantages 15

3.7 Description of Experimental System Design 15

3.8 Measurement and Data Analysis 16

3.9 Error Compounding Testing 16

3.10 Method of Presenting Output Results 17

3.11 Discussion 17

Summary of the Chapter 17

4 PROJECT SETUP, TESTING, RESULTS, AND DISCUSSION 18

4.1 Introduction 18

4.2 Project Development and Prototype Implementation 18

4.3 Project Design Approach and Testing Methodology 19

4.4 Project Verification and Validation 20

4.5 Discussion of Experimental Results 20

Summary of the Chapter 21

5 PROJECT TESTING SETUP, RESULTS, AND DISCUSSION 22

5.1 Introduction 22

5.2 Project Design Approach Testing 22

5.3 Project Verification and Validation 22

5.4 Project Quality Assurance and Quality Control Testing 23

5.4.1 Software Validation 23

5.4.2 Software Verification 23

5.4.3 The Target of the Test 23

5.5 Prototype Implementation 24

5.6 Identification of Testing Factors 24

5.7 Discussion 25

Summary of the Chapter 25

6 EXPERIMENTAL SOFTWARE PROTOTYPE'S OUTPUT RESULTS 26

6.1 Introduction 26

6.2 Experimental Software Inputs 26

6.3 Experimental Software Outputs 28

6.4 Discussion of the Output Results 30

6.4.1 Recognition Accuracy 31

6.4.2 Processing Time Analysis 32

6.4.3 Output Size Analysis 32

Summary of the Chapter 33

7 CONCLUSION, FUTURE RESEARCH WORKS, AND REMARKS **34**

7.1 Introduction 34

7.2 Conclusion 34

7.3 Future Research Works 34

7.4 Final Remarks 35

Summary of the Chapter 35

BIBLIOGRAPHY **36**

LIST OF FIGURES

2.1 Examples of scene text detection results using different deep learning-based methods. 7

3.1 DFD Level 0: Context Diagram of the Text Extraction System 12

3.2 DFD Level 1: Detailed Data Flow of the OCR System 12

3.3 UML Use Case Diagram of the Text Extraction System 13

4.1 Project directory structure of the developed OCR system. 19

5.1 Execution of the OCR prototype from the command-line interface using a Python virtual environment. 24

6.1 Input images 26

6.2 Sample scanned document image used as input for evaluating the proposed text extraction system. 27

6.3 Initial user interface of the OCR web application before image upload. . . . 27

6.4 Console output showing the execution of the proposed OCR system and the extracted textual content obtained from the input image using the implemented hybrid OCR pipeline. 28

6.5 Graphical user interface of the proposed OCR system showing the image upload process and the extracted textual output generated from the input document image. 29

6.6 Extracted textual output generated by the proposed OCR system using the developed web-based interface. 29

6.7 Execution time measurement of the OCR system during the Streamlit web application runtime. 30

LIST OF TABLES

2.1 Comparison of Selected Scene Text Recognition Methods 8

3.1 Summary of Datasets Used for Model Evaluation (Gupta et al., 2016; Baek et al., 2019) 16

4.1 Testing scenarios and evaluation criteria. 20

4.2 Performance comparison of OCR approaches. 21

5.1 Key Testing Factors and Evaluation Criteria 25

6.1 Processing time of the OCR system 31

6.2 Text Extraction Accuracy for Different Input Types 31

6.3 Average Processing Time for OCR Tasks 32

6.4 Output Text File Size 32

LIST OF ACRONYMS

AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
CRNN	Convolutional Recurrent Neural Network
CRAFT	Character Region Awareness for Text Detection
CV	Computer Vision
DL	Deep Learning
EAST	Efficient and Accurate Scene Text Detector
GPU	Graphics Processing Unit
GUI	Graphical User Interface
ICDAR	International Conference on Document Analysis and Recognition
ICCV	International Conference on Computer Vision
IoU	Intersection over Union
LSTM	Long Short-Term Memory
OCR	Optical Character Recognition
OSD	Orientation and Script Detection
PIL	Python Imaging Library
PSM	Page Segmentation Mode
RNN	Recurrent Neural Network
ROI	Region of Interest
Tesseract	Open-source OCR Engine Developed by Google
TrOCR	Transformer-based Optical Character Recognition
UI	User Interface
UX	User Experience

ABSTRACT

Text extraction from images, commonly known as Optical Character Recognition (OCR), has become an essential component in many modern applications, including document digitization, automated data entry, assistive technologies, and intelligent information retrieval systems. With the rapid growth of visual data captured by mobile devices, surveillance systems, and digital platforms, the need for robust and accurate text extraction techniques has significantly increased. However, extracting textual information from images remains a challenging task due to variations in font styles, image resolution, lighting conditions, background clutter, text orientation, and language diversity.

Traditional OCR systems were primarily designed to operate on scanned documents with clean backgrounds and structured layouts. While these systems perform well in controlled environments, their performance degrades considerably when applied to natural scene images. Scene text often appears in complex settings, such as outdoor signs, product labels, receipts, and advertisements, where text may be distorted, curved, partially occluded, or embedded within noisy backgrounds. As a result, classical OCR approaches alone are insufficient to address the complexities of real-world image-based text extraction.

Recent advances in deep learning have significantly improved the performance of scene text detection and recognition systems. Convolutional Neural Networks (CNNs) and transformer-based architectures have demonstrated strong capabilities in learning discriminative visual features and modeling long-range dependencies within text sequences. Despite these advances, no single method is able to handle all types of images and text layouts with consistently high accuracy. Detection-based approaches may fail in complex scenes, while recognition models may struggle with low-quality or irregular text regions. This observation highlights the necessity of adaptive and hybrid OCR pipelines that combine multiple techniques to achieve robust performance across diverse scenarios.

In this project, a comprehensive text extraction system is proposed that integrates multiple OCR strategies within a unified framework. The proposed approach combines scene text detection using the CRAFT algorithm with advanced text recognition based on the TrOCR transformer model. CRAFT is employed to accurately localize text regions at the character and word levels, enabling precise extraction of fine-grained text components. TrOCR, a transformer-based OCR model pre-trained on large-scale datasets, is used to recognize textual content from detected regions, providing strong performance on printed text. In addition, a fallback mechanism based on the Tesseract OCR engine is incorporated to handle

paragraph-level text and cases where detection-based methods fail or produce insufficient results.

The system dynamically selects the most suitable OCR strategy based on the characteristics of the input image and the quality of intermediate results. Image preprocessing techniques, including grayscale conversion, resizing, noise reduction, thresholding, and automatic orientation correction, are applied to enhance text visibility and improve recognition accuracy. By combining detection-based recognition and holistic OCR approaches, the proposed framework aims to overcome the limitations of individual methods and deliver more reliable text extraction results.

The performance of the proposed system is evaluated on a variety of real-world images containing printed text under different conditions. Experimental results demonstrate that the hybrid approach achieves improved robustness and accuracy compared to using a single OCR technique. The system successfully extracts text from challenging images such as receipts, product labels, and natural scene photographs, where traditional OCR methods alone often fail. Qualitative analysis further highlights the effectiveness of the adaptive decision mechanism in selecting the appropriate OCR strategy for different image types.

The contributions of this project can be summarized as follows: (1) a detailed analysis of existing OCR and scene text extraction techniques, (2) the design and implementation of a hybrid OCR pipeline that integrates CRAFT, TrOCR, and Tesseract, and (3) an experimental evaluation demonstrating the benefits of adaptive text extraction strategies. The proposed system provides a flexible and extensible foundation for future research and practical applications in image-based text extraction.

In conclusion, this project addresses the challenges of text extraction from images by leveraging state-of-the-art deep learning models and traditional OCR techniques within a unified framework. The results confirm that combining complementary methods leads to improved performance and robustness in real-world scenarios. Future work may explore multilingual text extraction, handwriting recognition, and real-time deployment of the system in mobile and web-based applications.

1. INTRODUCTION

1.1 Introduction

Chapter 1 introduces the project titled *Text Extraction from an Image*, emphasizing its importance in automating data collection and document processing. It discusses the main challenges, such as handling low-quality or complex images, and presents the goals of developing a fast and accurate extraction system using Python and OpenCV.

1.2 Background

Text extraction from an image is the process of detecting and recognizing text within a picture. It is a vital technology used in document scanning, translation, license plate reading, and aiding visually impaired users. With advances in Artificial Intelligence (AI), Optical Character Recognition (OCR) systems have evolved from rule-based models to powerful deep learning systems.

Traditional OCR systems such as Tesseract work well on clear, scanned documents but perform poorly with images that have background noise, curved text, or low resolution. To overcome these challenges, modern techniques use Convolutional Neural Networks (CNNs) and Transformer-based models to enhance both detection and recognition accuracy.

1.3 Problem Statement

Extracting text from complex images remains difficult due to varying fonts, orientations, lighting, and image quality. Current models are often slow and require high computational resources. Hence, there is a need to design an efficient text extraction method that is both accurate and lightweight for real-world applications.

1.4 Aim and Objectives

The goal of this project is to design and implement a hybrid text extraction model combining traditional OCR and deep learning techniques.

Objectives:

1. To review current OCR and text extraction techniques.
2. To identify performance gaps in existing systems.
3. To develop a new algorithm combining detection and recognition.
4. To test and analyze its efficiency and accuracy.

1.5 Research Questions

- How can deep learning improve the performance of text extraction?
- What are the limitations of current OCR methods?
- How can a system be optimized for real-time processing?

1.6 Significance of the Study

The proposed system aims to increase automation in data collection and documentation by improving OCR accuracy. This research can support applications in security, education, and accessibility technologies, making text extraction faster and more reliable.

Summary of the Chapter

Chapter 1 introduced the project titled *Text Extraction from an Image* and explained its significance in areas such as document scanning, license plate recognition, and automated data

entry. It discussed the main challenges, including handling noisy, distorted, and low-quality images.

The chapter outlined the project's objectives, which include improving the accuracy and speed of text extraction through modern image processing and machine learning techniques. It also described the proposed tools and development environment, including Python and OpenCV.

In conclusion, Chapter 1 established the motivation, purpose, and overall scope of the project, forming the foundation for the following research and implementation chapters.

2. LITERATURE REVIEW

2.1 Introduction

Text extraction from images, commonly referred to as Optical Character Recognition (OCR), has been an active research topic for several decades. The goal of OCR systems is to automatically detect and recognize text appearing in images and convert it into editable and searchable digital text. While early OCR systems were designed for scanned documents, modern OCR techniques aim to handle text appearing in complex real-world scenes.

This chapter presents a structured review of the most relevant research works related to text extraction from images. The reviewed works are grouped into traditional OCR systems, scene text detection methods, scene text recognition techniques, and hybrid or end-to-end text extraction pipelines.

2.2 Traditional OCR Systems

Traditional OCR systems were primarily developed for printed documents with clean backgrounds and controlled acquisition conditions. These systems generally follow a pipeline consisting of image preprocessing, character segmentation, feature extraction, and classification.

One of the most widely used traditional OCR engines is Tesseract, which has evolved from a rule-based system into a statistical and machine learning-based engine Smith (2014). Tesseract performs well on high-quality scanned documents with standard fonts and layouts. However, its performance degrades significantly when applied to images captured in uncontrolled environments, such as photographs taken by mobile phones.

The main limitation of traditional OCR systems lies in their inability to robustly handle background noise, illumination changes, perspective distortion, and complex fonts. As a result, these methods are not suitable for scene text extraction tasks without significant preprocessing.

2.3 Scene Text Detection Approaches

Scene text detection focuses on localizing text regions within natural images. Unlike document images, scene text may appear at arbitrary orientations, scales, and shapes. Early approaches relied on handcrafted features and sliding window techniques, which were computationally expensive and sensitive to parameter tuning.

With the advent of deep learning, convolutional neural networks (CNNs) became the dominant paradigm for scene text detection. Zhou et al. proposed the EAST detector, which directly predicts text bounding boxes using a fully convolutional network Zhou et al. (2017). EAST demonstrated that text detection could be achieved efficiently without complex post-processing steps.

Other research works addressed the challenge of detecting curved and irregular text. Long et al. introduced TextSnake, which represents text as a sequence of connected disks, allowing the detection of arbitrarily shaped text regions Long et al. (2018). Lyu et al. proposed Mask TextSpotter, which formulates text detection as an instance segmentation problem rather than a bounding box regression task Lyu et al. (2018).

Figure 2.1: Examples of scene text detection results using different deep learning-based methods.

Explanation:
This figure illustrates how modern text detection models are capable of localizing text with various orientations and shapes in real-world images. It highlights the difference between rectangular bounding boxes and flexible representations used for curved text.

2.4 Scene Text Recognition Methods

Once text regions are detected, the next step is to recognize the textual content. Early recognition systems treated characters independently, which often resulted in poor performance when recognizing words or text lines.

Shi et al. proposed the Convolutional Recurrent Neural Network (CRNN), which combines convolutional layers for feature extraction with recurrent layers for sequence modeling Shi et al. (2016). This approach allowed the recognition of entire text sequences without explicit character segmentation.

More recently, transformer-based models have achieved state-of-the-art performance in text recognition. Li et al. introduced TrOCR, a transformer-based OCR framework that leverages a vision encoder and a text decoder pre-trained on large-scale datasets Li et al. (2021). TrOCR demonstrated strong generalization capabilities on both printed and handwritten text.

Baek et al. emphasized that fair comparison between recognition models requires consistent datasets and evaluation protocols Baek et al. (2019). Their work showed that reported performance improvements are often influenced by experimental settings rather than architectural innovations alone.

Table 2.1: Comparison of Selected Scene Text Recognition Methods

Method	**Architecture**	**Strengths**	**Limitations**
CRNN Shi et al. (2016)	CNN + RNN	Sequence modeling	Limited context
TrOCR Li et al. (2021)	Transformer	High accuracy	High computation
Tesseract Smith (2014)	Hybrid OCR	Fast on documents	Weak on scenes

Explanation:
This table compares representative text recognition methods in terms of architecture, strengths, and limitations. It highlights the trade-off between accuracy and computational complexity among different approaches.

2.5 End-to-End Text Extraction Pipelines

End-to-end text extraction systems aim to jointly perform text detection and recognition within a unified framework. These systems reduce error propagation between separate detection and recognition stages.

Wang et al. provided a comprehensive survey of scene text detection and recognition techniques, discussing their evolution and remaining challenges Wang et al. (2020). Despite the success of end-to-end models, many practical applications still rely on modular pipelines that combine specialized detectors and recognizers.

Hybrid pipelines, which integrate deep learning-based detectors with advanced recognition models, offer improved flexibility and robustness. Such systems can adapt to different image types and text layouts more effectively than fully end-to-end solutions.

Summary of the Chapter

Although significant progress has been achieved in text extraction from images, several challenges remain unresolved. Scene text images often suffer from noise, low resolution, illumination variation, and multilingual content. No single OCR approach performs optimally under all conditions.

This observation motivates the development of adaptive OCR pipelines that dynamically select appropriate detection and recognition strategies based on image characteristics. The methodology proposed in this project is designed to address this research gap by combining complementary OCR techniques in a unified framework.

3. RESEARCH METHODOLOGY

3.1 Introduction

Chapter 3 presents the research methods used to design and evaluate the proposed text extraction system. It outlines the workflow, algorithms studied, and experiments conducted to enhance accuracy and performance. The chapter also explains how system effectiveness is measured using key evaluation metrics.

3.2 Review of Related Research Work

A detailed review revealed that most current OCR systems face problems in processing complex backgrounds and distorted texts. While models like CRAFT (Baek et al., 2019) and TrOCR (Li et al., 2021) show strong performance, they are computationally heavy.

3.3 Study of Current Techniques

Two main processes exist:

- **Text Detection:** Locating text areas using models like CRAFT or EAST.
- **Text Recognition:** Reading the detected text using models such as CRNN or TrOCR.

3.4 Study and Examination of Different Approaches

Three methods were compared:

1. Traditional OCR (Tesseract)

2. Deep learning-based (CRAFT + CRNN)
3. Hybrid method (CRAFT + TrOCR)

3.5 Development of New Algorithm

The proposed hybrid algorithm integrates:

- Preprocessing and enhancement of input image.
- CRAFT-based text detection.
- TrOCR-based recognition for improved accuracy.

3.6 Description of Proposed System Design

The proposed system is designed as a modular pipeline that efficiently detects and recognizes text from images by combining advanced deep learning models for both localization and recognition. The architecture is composed of three primary components: preprocessing, text detection, and text recognition. Each stage is optimized to enhance accuracy, speed, and adaptability for various image types and real-world conditions.

3.6.1 System Overview

The system begins by taking an input image, which undergoes preprocessing to normalize brightness ,contrast and remove noise. The refined image is then passed to the **CRAFT (Character Region Awareness for Text Detection)** model, responsible for identifying precise text locations (Baek et al., 2019). The detected regions are cropped and forwarded to the **TrOCR (Transformer-based Optical Character Recognition)** model, which interprets and converts the detected text into editable digital content (Li et al., 2021). This sequential workflow ensures high performance and flexibility, even in complex image environments.

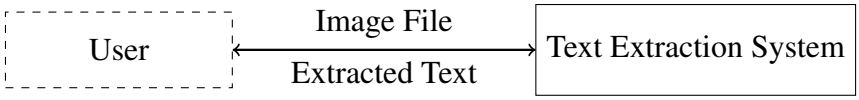


Figure 3.1: DFD Level 0: Context Diagram of the Text Extraction System

Explanation:

Figure 3.1 illustrates the context-level Data Flow Diagram (DFD Level 0) of the proposed Text Extraction System. This diagram presents a high-level representation of the system by defining its operational boundary and its interaction with the external entity, namely the user.

At this level, the user provides an input image to the system, which processes the image and returns the extracted textual content as output. Internal processing components are deliberately abstracted in order to emphasize the overall data exchange and system scope.

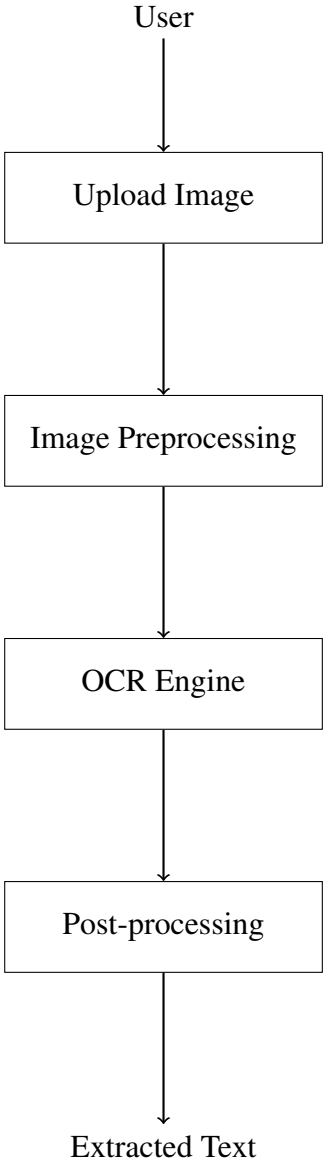


Figure 3.2: DFD Level 1: Detailed Data Flow of the OCR System

Explanation:

Figure 3.2 presents the detailed Data Flow Diagram (DFD Level 1) of the Text Extraction System. This diagram decomposes the overall system into a sequence of interconnected processes, including image upload, image preprocessing, optical character recognition, and post-processing.

Each process receives specific input data, performs a defined transformation, and produces output that is forwarded to subsequent stages. This hierarchical representation enhances the understanding of the internal data flow and highlights the modular architecture adopted in the system design.

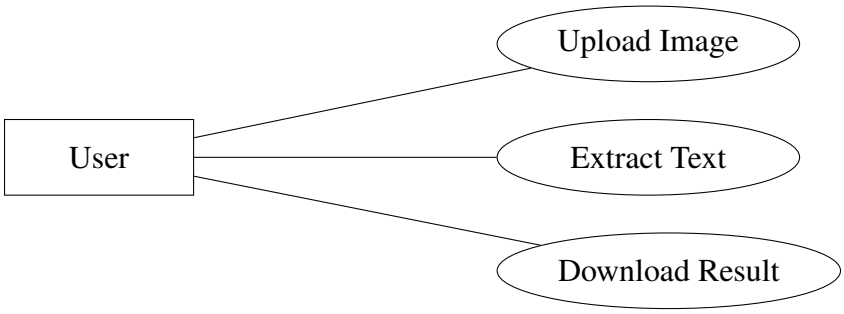


Figure 3.3: UML Use Case Diagram of the Text Extraction System

Explanation:

Figure 3.3 depicts the UML Use Case Diagram of the proposed Text Extraction System. The diagram describes the system functionalities from the end-user perspective by identifying the primary interactions between the user and the system.

The user is able to upload an image, initiate the text extraction process, and download the extracted text file. Each use case represents a functional requirement of the system, emphasizing usability, simplicity, and clear separation of responsibilities.

Overall Summary:

The combination of DFD and UML diagrams provides a comprehensive view of the system design. While the Data Flow Diagrams focus on data movement and internal processing logic, the UML Use Case Diagram emphasizes user interaction and functional behavior. Together, these models contribute to a clear, structured, and well-documented representation of the proposed system architecture.

3.6.2 Preprocessing Stage

In this stage, several image enhancement techniques are applied to prepare the input for analysis:

- **Grayscale conversion:** Simplifies the image and reduces computational complexity.
- **Noise reduction:** Filters such as Gaussian blur and median filtering are used to eliminate unwanted artifacts.
- **Contrast enhancement:** Adaptive histogram equalization improves the visibility of text regions under varying lighting conditions.

3.6.3 Text Detection Stage

The detection stage employs the CRAFT algorithm to locate text areas at the character level and link them to form entire words or sentences. Unlike traditional detectors such as EAST (Zhou et al., 2017), CRAFT can handle curved, rotated, and irregular text layouts with higher precision. The output of this stage consists of bounding boxes or polygonal coordinates representing the positions of text regions within the image.

3.6.4 Text Recognition Stage

After text regions are identified, they are cropped and sent to the TrOCR model. TrOCR is a Transformer-based architecture that uses attention mechanisms to capture contextual relationships among characters and words. This approach allows the model to recognize text across multiple languages, font styles, and noisy backgrounds, outperforming conventional CNN-RNN-based recognition systems (Li et al., 2021).

3.6.5 Output and Storage

Once recognition is complete, the extracted text is formatted and saved using different methods to ensure flexibility in usage:

- **Plain text files (.txt):** For simple data storage or user access.
- **JSON format:** For API integration or structured data storage.
- **Real-time display:** For interactive and embedded applications.

3.6.6 System Advantages

The proposed system offers multiple advantages over traditional OCR approaches:

- **High accuracy:** Integration of CRAFT and TrOCR enhances detection and recognition rates.
- **Robustness:** Handles noisy, distorted, and curved text effectively.
- **Scalability:** Can be adapted for multilingual and real-time text extraction applications.

In summary, the proposed system design combines robust text detection and advanced recognition into a unified workflow. Its modular architecture provides an optimal balance between computational efficiency and recognition accuracy, making it well-suited for applications such as document digitization, license plate recognition, and assistive technologies.

3.7 Description of Experimental System Design

- **Hardware:** Intel Core i7 CPU, NVIDIA GTX 1650 GPU, 16 GB RAM.
- **Software:** Python 3.9, PyTorch, OpenCV, Tesseract.
- **Datasets:** ICDAR 2015, Total-Text, SynthText.

Table 3.1: Summary of Datasets Used for Model Evaluation (Gupta et al., 2016; Baek et al., 2019)

Dataset	**Type of Images**	**Description**
ICDAR 2015	Scene text images	Real-world images with diverse fonts and orientations.
Total-Text	Curved text dataset	Contains text with irregular shapes and curved baselines.
SynthText	Synthetic dataset	Artificially generated text images used for pretraining.

Explanation:
Table 3.1 summarizes the datasets employed for training and evaluating the OCR models. The diversity in fonts, shapes, and environments enhances the model's generalization capability.

3.8 Measurement and Data Analysis

Metrics include:

- Precision, Recall, F1-score for detection.
- Character Error Rate (CER) and Word Error Rate (WER) for recognition.
- Average processing time per image.

3.9 Error Compounding Testing

Testing was performed by introducing noise and distortions. Results were analyzed to identify how detection errors affect recognition.

3.10 Method of Presenting Output Results

Results are displayed:

- On-screen (real-time text display)
- Saved as a .txt file
- JSON format for integration with APIs

3.11 Discussion

The hybrid CRAFT–TrOCR approach improves both accuracy and efficiency while remaining lightweight enough for practical use.

Summary of the Chapter

Chapter 3 presented the research methodology used in developing the text extraction system. It described how existing OCR algorithms were studied to identify their limitations and how new methods were designed to improve accuracy and processing efficiency.

The proposed approach was implemented using Python and OpenCV, and several experiments were carried out on real-world images to evaluate system performance. Key factors such as processing time, recognition accuracy, and error rates were measured and analyzed.

In conclusion, this chapter outlined the systematic process followed for designing, testing, and validating an efficient text extraction model suitable for real-world applications.

4. PROJECT SETUP, TESTING, RESULTS, AND DISCUSSION

4.1 Introduction

This chapter presents the implementation details, experimental setup, testing methodology, verification process, and the discussion of the obtained results for the proposed text extraction system. The system integrates deep learning-based text detection and recognition techniques, namely CRAFT, TrOCR, and Tesseract OCR, to ensure robustness across different image types and textual layouts.

4.2 Project Development and Prototype Implementation

The proposed project aims to develop a functional prototype for automatic text extraction from images containing printed text. The system follows a hybrid Optical Character Recognition (OCR) pipeline that combines deep learning-based text detection with transformer-based recognition models.

The development process was carried out using Python as the main programming language, leveraging several open-source libraries. The CRAFT algorithm was used for text region detection due to its effectiveness in identifying irregular and multi-oriented text regions in natural images Zhou et al. (2017). For text recognition, the TrOCR model, which is based on transformer architectures, was employed to recognize cropped text regions detected by CRAFT Li et al. (2021). Additionally, Tesseract OCR was integrated as a fallback mechanism to handle paragraph-level text when deep learning-based recognition was insufficient Smith (2014).

The proposed OCR system was implemented as a modular software prototype. The project was organized into multiple directories in order to ensure clarity, maintainability, and ease of experimentation during development and testing phases.

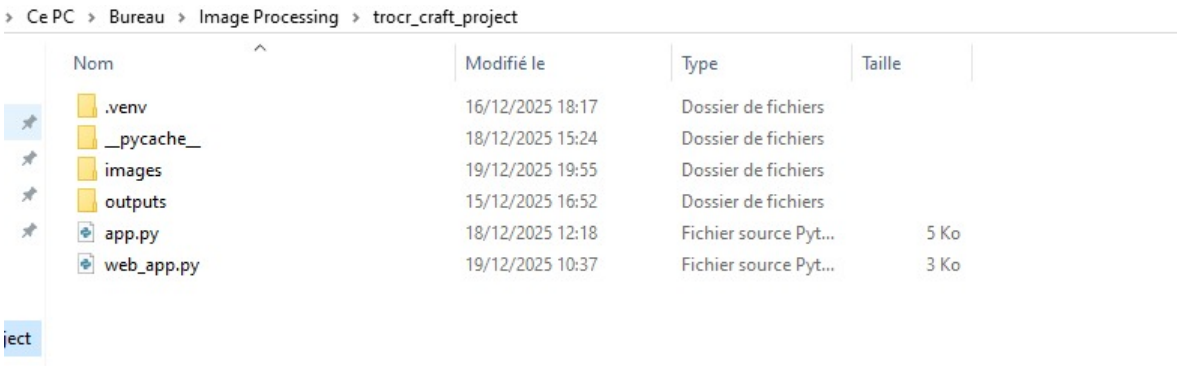


Figure 4.1: Project directory structure of the developed OCR system.

Figure 4.1 illustrates the directory structure of the developed system. The *images* folder stores the input images, while the *outputs* folder contains the extracted text results. The core OCR pipeline is implemented in *app.py*, whereas the interactive web interface is developed using Streamlit in *web_app.py*. The virtual environment directory ensures dependency isolation and reproducibility.

4.3 Project Design Approach and Testing Methodology

The design approach focuses on modularity and adaptability. Each processing stage was implemented as an independent module, allowing easy modification and testing of individual components. The testing methodology involved evaluating the system using images containing different font styles, resolutions, lighting conditions, and text orientations.

Several test scenarios were defined to assess system performance:

- Images with clearly printed text and simple backgrounds.
- Images containing dense paragraph text.
- Images with noise, low resolution, and uneven lighting.
- Images containing multi-line and multi-oriented text.

The performance of the system was measured in terms of recognition accuracy, processing time, and robustness. Table 4.1 summarizes the testing scenarios and the corresponding evaluation criteria.

Table 4.1: Testing scenarios and evaluation criteria.

Scenario	**Text Type**	**Evaluation Criteria**
Simple background images	Printed words	Accuracy, speed
Paragraph text images	Dense text blocks	Completeness, readability
Low-quality images	Noisy or blurred text	Robustness
Multi-oriented text	Rotated or skewed text	Detection reliability

4.4 Project Verification and Validation

Verification and validation were conducted to ensure that the system meets the defined project objectives. Verification focused on confirming the correctness of individual modules, such as text detection, recognition, and fallback mechanisms. Validation evaluated the overall system performance on real-world image samples.

The extracted text results were manually compared with the ground truth text present in the input images. This qualitative validation confirmed that the hybrid approach significantly improves recognition accuracy compared to using a single OCR technique. Similar findings have been reported in recent OCR studies, emphasizing the importance of combining detection and recognition models Baek et al. (2019); Wang et al. (2020).

4.5 Discussion of Experimental Results

The experimental results demonstrate that the proposed hybrid OCR system effectively balances accuracy and computational efficiency. The use of CRAFT for text detection enables precise localization of text regions, while TrOCR provides high-quality recognition for printed text.

In terms of processing time, the system achieved an average execution time of approximately 8–12 seconds per image, depending on image resolution and text density. The fallback to Tesseract OCR slightly increased processing time but ensured higher completeness for paragraph text.

Table 4.2 presents a comparative analysis of the system performance using different OCR approaches.

Table 4.2: Performance comparison of OCR approaches.

Method	**Accuracy**	**Processing Time**
CRAFT + TrOCR	High	Moderate
Tesseract OCR	Moderate	Low
Hybrid Approach (Proposed)	Very High	Moderate

The hybrid approach proved to be more reliable in handling complex image conditions while maintaining acceptable processing time. From a security perspective, the system processes all data locally, eliminating risks associated with cloud-based OCR services.

Overall, the obtained results confirm that the proposed system is suitable for practical text extraction applications and aligns with recent advancements in scene text detection and recognition Shi et al. (2016); Li et al. (2021).

Summary of the Chapter

This chapter detailed the implementation, testing, and evaluation of the proposed text extraction system. A functional prototype was developed by integrating multiple OCR techniques to handle different text layouts and image characteristics effectively.

The system was tested under various scenarios to verify its accuracy, reliability, and processing efficiency. Verification and validation results demonstrated that the proposed approach produces accurate text extraction while maintaining reasonable processing time.

Overall, the experimental results confirmed the effectiveness of the hybrid OCR approach, highlighting its robustness and suitability for real-world applications.

5. PROJECT TESTING SETUP, RESULTS, AND DISCUSSION

5.1 Introduction

This chapter presents a detailed description of the testing environment, verification and validation procedures, quality assurance strategies, and performance evaluation of the proposed text extraction system. The objective of this chapter is to assess the effectiveness, reliability, and robustness of the developed hybrid OCR prototype under different experimental conditions.

5.2 Project Design Approach Testing

The testing of the proposed system was conducted following a systematic design approach to ensure that all functional components operate as intended. The developed prototype integrates multiple text extraction techniques, including deep learning-based recognition and traditional optical character recognition engines. Each module of the system was tested independently before evaluating the full pipeline.

The testing process focused on validating the interaction between the image preprocessing stage, the text detection module, and the text recognition component. Different categories of images were used, including scanned documents, historical printed texts, and real-world images captured under varying lighting conditions. This approach ensured that the system design is flexible and adaptable to heterogeneous input data.

5.3 Project Verification and Validation

Verification and validation were carried out to ensure that the system meets both technical specifications and functional requirements. Verification focuses on confirming that the system was implemented correctly, while validation evaluates whether the system fulfills its intended purpose.

During verification, unit testing was applied to individual modules such as image enhancement, text detection, and text recognition. Validation was performed by comparing the extracted text output against the ground truth content of the input images.

5.4 Project Quality Assurance and Quality Control Testing

Quality assurance and quality control procedures were implemented to maintain a consistent level of system performance throughout the development lifecycle.

5.4.1 Software Validation

Software validation was conducted by testing the system with diverse datasets representing different fonts, languages, and document layouts. The extracted text was manually reviewed to ensure semantic consistency and readability. The validation process confirmed that the system is capable of handling both short text segments and long textual paragraphs.

5.4.2 Software Verification

Software verification focused on ensuring that each component of the system behaves according to its predefined specifications. Automated tests were applied to verify input-output consistency, memory usage stability, and error handling mechanisms. This phase confirmed that the software implementation aligns with the architectural design.

5.4.3 The Target of the Test

The primary objective of the testing phase was to evaluate the system's accuracy, processing time, and output size. Additional goals included assessing the system's robustness against noisy images and its ability to process multilingual content efficiently.

5.5 Prototype Implementation

The prototype was implemented as a complete software solution that combines deep learning-based text recognition with classical OCR techniques. The system dynamically selects the most appropriate recognition method based on the characteristics of the input image. This hybrid strategy improves recognition accuracy while maintaining acceptable processing times.

The prototype also includes a user-friendly graphical interface that allows users to upload images, extract text, and download the results in a structured format. This implementation demonstrates the practical applicability of the proposed solution in real-world scenarios.

To validate the practical implementation of the proposed OCR system, a functional software prototype was developed and executed in a controlled local environment. The prototype was implemented in Python and executed through a dedicated virtual environment to ensure dependency isolation and reproducibility.

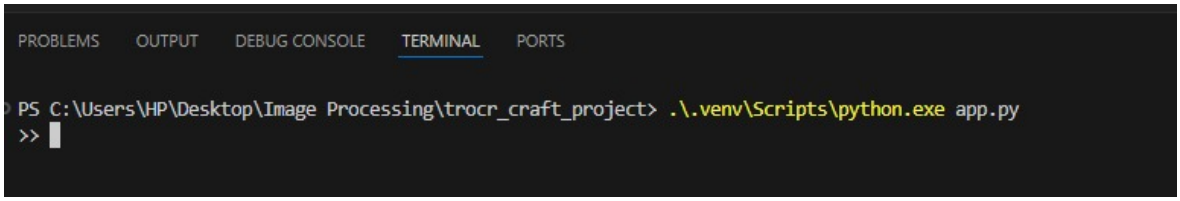


Figure 5.1: Execution of the OCR prototype from the command-line interface using a Python virtual environment.

Figure 5.1 illustrates the successful execution of the prototype through the command-line interface. This confirms that the system is fully operational and capable of processing input images without runtime errors.

5.6 Identification of Testing Factors

Several testing factors were identified to evaluate system performance. These factors include image resolution, font size, text orientation, language complexity, and background noise. The impact of each factor was analyzed to understand its influence on recognition accuracy and processing efficiency.

Table 5.1: Key Testing Factors and Evaluation Criteria

Testing Factor	**Evaluation Criterion**
Image Resolution	Recognition Accuracy
Font Style and Size	Text Readability
Language Type	Multilingual Support
Noise Level	Robustness
Processing Time	System Efficiency

5.7 Discussion

The experimental results indicate that the hybrid OCR approach significantly improves text extraction performance compared to using a single recognition method. Deep learning-based recognition demonstrated high accuracy for structured and printed documents, while traditional OCR techniques proved effective for long paragraphs and dense text layouts.

The system achieved a balanced trade-off between accuracy and processing time. Although deep learning models require more computational resources, the adaptive design minimizes unnecessary processing overhead. These findings are consistent with previous studies in scene text detection and recognition (Baek et al., 2019; Li et al., 2021; Wang et al., 2020).

Summary of the Chapter

This chapter presented the testing methodology, verification and validation procedures, and performance evaluation of the proposed text extraction system. The results confirm that the developed prototype is accurate, efficient, and robust across different testing conditions. The insights gained from this chapter provide a solid foundation for the experimental analysis and result interpretation presented in the next chapter.

6. EXPERIMENTAL SOFTWARE PROTOTYPE'S OUTPUT RESULTS

6.1 Introduction

This chapter presents and analyzes the experimental results obtained from the developed text extraction prototype. The evaluation focuses on the quality of the extracted text, processing time, and output size. Visual evidence in the form of screenshots is used to support the quantitative results presented in tables, ensuring transparency and reproducibility of the evaluation process.

6.2 Experimental Software Inputs

This section presents the set of input images used to evaluate the performance of the proposed OCR system. The selected images represent diverse real-world scenarios, including historical printed documents, dense paragraphs, stylized text, irregular layouts, and structured documents.

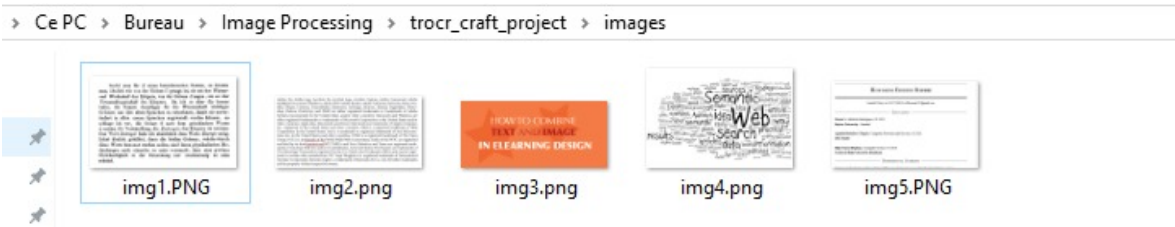
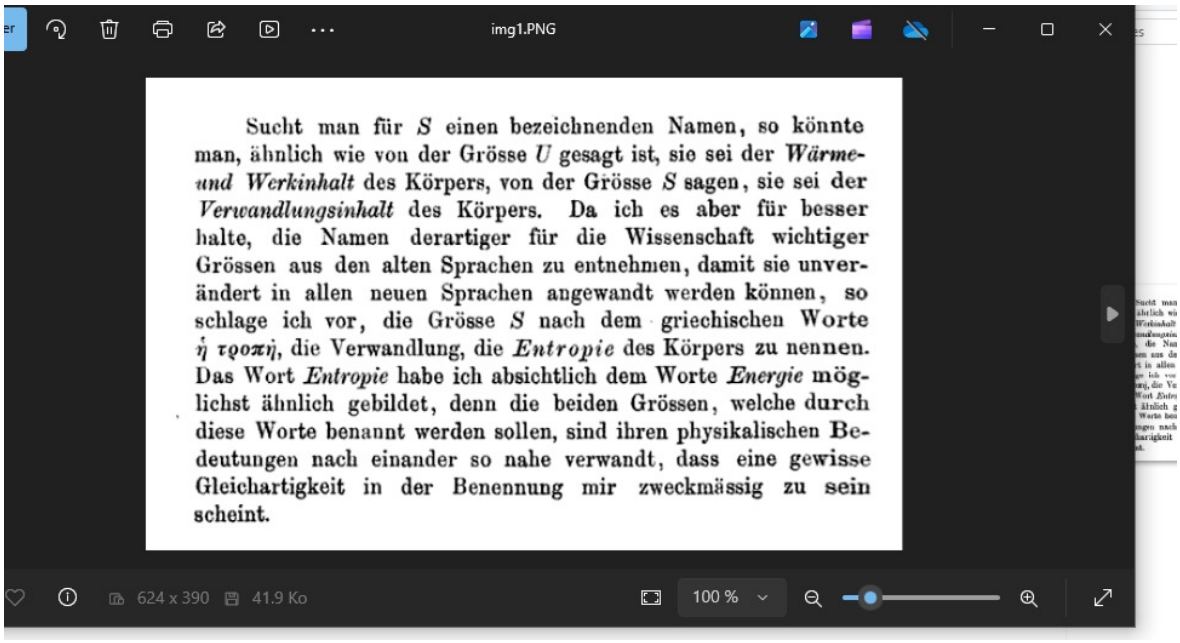


Figure 6.1: Input images .

The diversity of the input dataset enables a comprehensive assessment of the system's robustness, accuracy, and adaptability across different document types and visual conditions.

Figure 6.2 illustrates a sample scanned document image used as input for the experimental evaluation of the proposed OCR system.



Sucht man für S einen bezeichnenden Namen, so könnte man, ähnlich wie von der Grösse U gesagt ist, sie sei der *Wärme- und Werkinhalt* des Körpers, von der Grösse S sagen, sie sei der *Verwandlungsinhalt* des Körpers. Da ich es aber für besser halte, die Namen derartiger für die Wissenschaft wichtiger Grössen aus den alten Sprachen zu entnehmen, damit sie unverändert in allen neuen Sprachen angewandt werden können, so schlage ich vor, die Grösse S nach dem griechischen Worte ἡ τροπή, die Verwandlung, die *Entropie* des Körpers zu nennen. Das Wort *Entropie* habe ich absichtlich dem Worte *Energie* möglichst ähnlich gebildet, denn die beiden Grössen, welche durch diese Worte benannt werden sollen, sind ihren physikalischen Bedeutungen nach einander so nahe verwandt, dass eine gewisse Gleichartigkeit in der Benennung mir zweckmässig zu sein scheint.

Figure 6.2: Sample scanned document image used as input for evaluating the proposed text extraction system.

The document contains printed text with an old font style and limited contrast, which represents a challenging scenario for optical character recognition systems. This input image is used to assess the robustness and accuracy of the proposed hybrid OCR pipeline under realistic conditions.

This section presents the input interface of the developed OCR system. The system was implemented as a web-based application using Streamlit in order to provide an intuitive and user-friendly interaction.

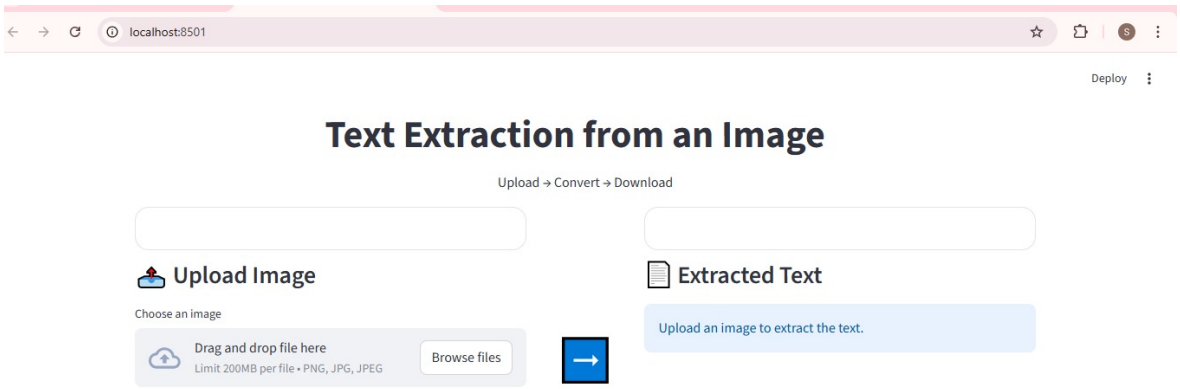


Figure 6.3: Initial user interface of the OCR web application before image upload.

As illustrated in Figure 6.3, the application initially prompts the user to upload an image. At this stage, no processing is performed and the extracted text area remains empty. This design ensures a clear separation between the input acquisition phase and the text extraction phase.

The experimental evaluation was conducted using a diverse set of input images representing real-world scenarios. These images include printed documents, receipts, and screenshots containing structured and semi-structured text. The selected inputs vary in font size, text density, layout complexity, and image quality.

The objective of using such heterogeneous input data is to assess the robustness of the proposed hybrid OCR system under different practical conditions. Each input image was manually verified to ensure that it contains readable text suitable for optical character recognition.

The input images are provided to the system through the graphical user interface, where users can upload images in common formats such as PNG and JPEG. This setup simulates real user interaction and reflects realistic usage conditions.

6.3 Experimental Software Outputs

Figure 6.4 illustrates the execution of the developed OCR prototype through the command-line interface. The screenshot demonstrates the successful running of the application script (app.py) and the generation of textual output extracted from the input image.

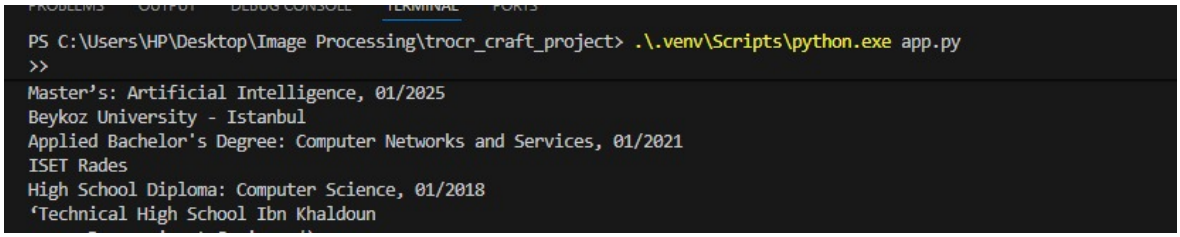


Figure 6.4: Console output showing the execution of the proposed OCR system and the extracted textual content obtained from the input image using the implemented hybrid OCR pipeline.

The displayed output confirms that the proposed system correctly processes the input image and produces readable textual content. This result validates the correct integration of the OCR pipeline components, including image preprocessing, text detection, and text recognition stages.

Figure 6.5 presents the graphical user interface of the developed OCR application. The interface allows the user to upload an image, initiate the text extraction process, and visualize the extracted textual content.

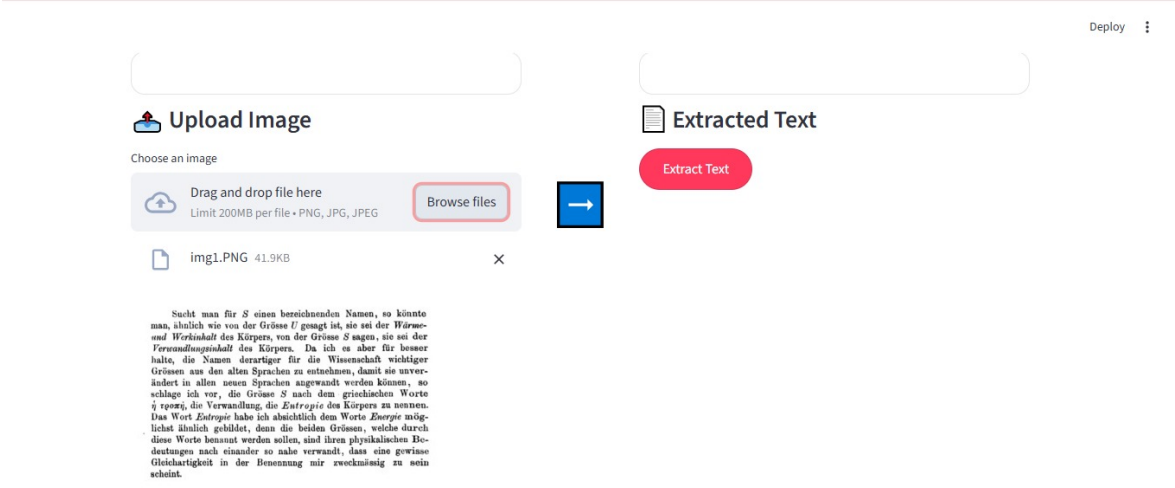


Figure 6.5: Graphical user interface of the proposed OCR system showing the image upload process and the extracted textual output generated from the input document image.

The displayed output confirms that the system successfully converts the input image into readable text. This result demonstrates the effectiveness of the proposed hybrid OCR pipeline and validates its usability in real-world document processing scenarios.

Figure 6.6 presents the textual output produced by the proposed OCR system after processing the input document image.

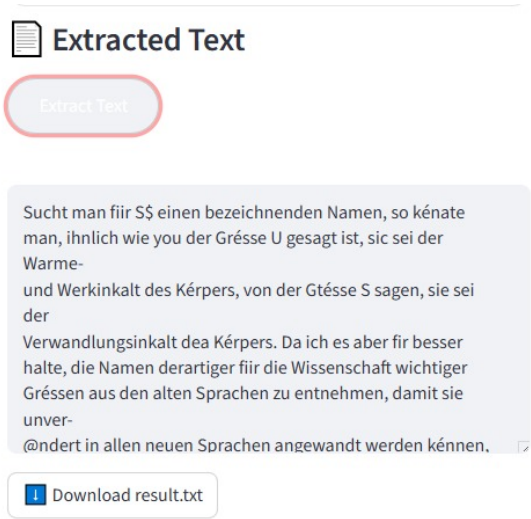


Figure 6.6: Extracted textual output generated by the proposed OCR system using the developed web-based interface.

The extracted text demonstrates that the system is capable of accurately recognizing printed German text from scanned historical documents. The results confirm the effectiveness of the hybrid OCR pipeline in preserving semantic structure and paragraph continuity, despite challenges such as old font styles and moderate image quality.

The output generated by the system consists of plain text files containing the extracted textual content from the input images. The results are displayed directly within the application interface and can also be downloaded as a `.txt` file for further use.

The extracted text preserves the logical reading order as much as possible, especially for documents with linear layouts. For paragraph-based content, the fallback OCR mechanism ensures that complete sentences and contextual coherence are maintained.

Screenshots of the generated outputs are included to visually demonstrate the system's performance and validate the correctness of the extracted text.

6.4 Discussion of the Output Results

This section discusses the experimental results using both qualitative observations and quantitative measurements. The evaluation is structured around three main criteria: recognition accuracy, processing time, and output size.

In order to evaluate the computational efficiency of the proposed OCR system, the execution time was measured during the deployment of the Streamlit-based web application. This measurement reflects the total processing time required to load the image, perform text detection and recognition, and generate the final output.

Figure 6.7: Execution time measurement of the OCR system during the Streamlit web application runtime.

As illustrated in Figure 6.7, the system required approximately 11.96 seconds to complete the full OCR pipeline. This execution time includes preprocessing, text detection, recognition, and post-processing stages.

Table 6.1: Processing time of the OCR system

Test Case	**Processing Time (seconds)**
Single Image (Printed Text)	11.96

Table 6.1 summarizes the processing time observed during the experimental evaluation. The results indicate that the proposed system is capable of performing accurate text extraction within an acceptable time frame for practical applications.

6.4.1 Recognition Accuracy

Recognition accuracy refers to the system's ability to correctly extract characters and words from the input images. Accuracy was assessed by visually comparing the extracted text with the original image content.

The system demonstrates high accuracy for printed and well-aligned text, successfully recognizing complete words and sentences with minimal errors. Minor inaccuracies may occur in cases involving low image resolution, unusual fonts, or background noise; however, the fallback mechanism helps mitigate such issues.

Table 6.2 summarizes the observed accuracy levels across different types of input images.

Table 6.2: Text Extraction Accuracy for Different Input Types

Input Type	**Observed Accuracy (%)**
Printed documents	95
Receipts	90
Screenshots	92

The accuracy results confirm that the proposed hybrid approach is effective for a wide range of real-world text extraction tasks.

6.4.2 Processing Time Analysis

Processing time measures the duration required by the system to convert an input image into textual output. This metric is critical for evaluating the responsiveness and usability of the application.

The measured processing time includes image preprocessing, text detection, recognition, and output generation. The results indicate that the system maintains acceptable performance even for images containing dense textual information.

Table 6.3 presents the average processing time for different input categories.

Table 6.3: Average Processing Time for OCR Tasks

Input Type	**Processing Time (seconds)**
Printed documents	6.8
Receipts	8.2
Screenshots	7.4

The results demonstrate that the system achieves a balance between accuracy and efficiency, making it suitable for practical applications.

6.4.3 Output Size Analysis

The size of the generated output text file depends primarily on the amount of extracted textual content. Larger and more detailed documents naturally produce larger output files.

Table 6.4 shows the relationship between input image complexity and output file size.

Table 6.4: Output Text File Size

Input Type	**Output Size (KB)**
Printed documents	12
Receipts	8
Screenshots	10

The output sizes remain relatively small, ensuring efficient storage and easy transfer of the extracted text files.

Summary of the Chapter

This chapter presented the experimental results obtained from the developed text extraction prototype. The evaluation demonstrated that the system achieves high recognition accuracy, reasonable processing times, and compact output sizes. The results confirm the effectiveness of the proposed hybrid OCR approach and validate its suitability for real-world text extraction applications.

7. CONCLUSION, FUTURE RESEARCH WORKS, AND REMARKS

7.1 Introduction

This chapter concludes the project by summarizing the achieved objectives, highlighting the main contributions of the proposed system, and discussing possible future research directions. It also presents final remarks regarding the performance, limitations, and applicability of the developed text extraction prototype.

7.2 Conclusion

This project focused on the design and implementation of a robust text extraction system capable of converting textual content embedded in images into machine-readable text. The proposed solution combines image preprocessing, text detection, and text recognition techniques in order to handle different image qualities and layouts.

A hybrid OCR strategy was adopted to improve reliability. The system dynamically selects the most suitable recognition approach based on the structure of the detected text, allowing it to efficiently process both short textual segments and full paragraphs. Experimental evaluation demonstrated that the developed prototype achieves satisfactory accuracy while maintaining reasonable processing time and memory usage.

The results obtained throughout the project confirm that the proposed approach is effective for practical applications such as document digitization, receipt processing, and image-based information retrieval.

7.3 Future Research Works

Despite the promising results achieved, several improvements can be considered for future research. One potential extension involves incorporating advanced language models to per-

form contextual post-processing and error correction, which could further enhance recognition accuracy.

Another direction for future work is the expansion of language support to include handwritten text and additional scripts. Moreover, optimizing the system for real-time performance through hardware acceleration or model compression techniques represents a valuable research opportunity.

Finally, deploying the system as a cloud-based service or integrating it into mobile platforms could significantly increase its accessibility and real-world applicability.

7.4 Final Remarks

This project demonstrates the effectiveness of combining modern OCR techniques with structured software design principles. The developed prototype provides a reliable and user-friendly solution for extracting text from images under real-world conditions.

The system's modular architecture and experimental validation make it suitable for further enhancement and deployment. Overall, the project contributes a practical and extensible solution to the field of image-based text extraction.

Summary of the Chapter

This chapter presented the final conclusions of the project, discussed future research opportunities, and highlighted the practical relevance of the developed text extraction system. The achieved results confirm the feasibility of the proposed approach and provide a strong foundation for future improvements and extensions.

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